<table>
<tr><td>19.</td><td>(a)
Palindromic sequences
Restriction endonuclease / EcoRI.
(b)
Restriction endonucleases cut the strands of DNA a little away from the centre of the palindrome sites to form sticky end on each strand / Restriction endonucleases cut at specific recognition site between two same bases in opposite strands to produce sticky ends / EcoRI cuts the DNA between bases G and A in both the strand in the sequence
5'-GAATTC-3'
3'-CTTAAG-5'</td><td>½
½

1</td><td>2</td></tr>
<tr><td>20.</td><td>(a)
Case 1 –Male heterogamety
Organism – Many insects / grasshopper / any correct example.
Case II–Female heterogamety
Organism – Birds / hen / any correct example.

OR

(b)
(i)<table><tr><td>DNA nucleotide</td><td>RNA nucleotide</td></tr><tr><td>Has deoxyribose sugar</td><td>Has Ribose sugar</td></tr><tr><td>Types of nitrogen bases are A,G,C,T</td><td>Types of nitrogen bases are A,G,C,U</td></tr></table>(Any one difference)
(ii) Bonds in a nucleotide
N – Glycosidic linkage (between Sugar and Base)
Phosphoester linkage – (Between Sugar and Phosphate)</td><td>½
½
½
½

1

½
½</td><td>2</td></tr>
<tr><td>21.</td><td>(a)</td><td></td><td></td></tr>
</table>

(i) • Inverted Pyramid of Biomass	½	
• Such pyramids are seen in- Aquatic Conditions where a small standing crop of phytoplanktons supports a large standing crop of zooplankton or fish / In terrestrial ecosystem – when a large number of insects are feeding on a single tree / or any other correct example	½	
(ii) -Ecological pyramids does not take into account the same species belonging to two or more trophic levels. -It does not accommodate a food web. -Saprophytes are not given any place in ecological pyramids even though they play a vital role in the ecosystem. **(Any two)**	½+½	
OR		
(b) (i) -Decline in plant production. -Lowered resistance to environmental perturbations as drought. -Increased variability in certain ecosystem processes such as water use and pest and disease cycles - Any other correct point **(Any two points)**	½+½	
(ii) -Habitat loss and fragmentation. -Alien species invasion. -Over exploitation -Co-extinction **(Any two)**	½+½	2